THE QUOTIENT BY THE POWER-TORSION SUBMODULE HAS NO I-TORSION, AND POWER-TORSION MODULES ARE CLOSED UNDER EXTENSIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The counted statement span writes the power-torsion submodule using the union symbol in the exponent, delivered verbatim. The single reference inside the counted proof span is recorded as a non-load-bearing navigation pointer.

Allowed standard prerequisites (editorial).

- The I-power torsion submodule M[I^infinity] of a module, as the union of the M[I^n]. used at statement lines 24045-24046 and proof lines 24051-24054. Is the object whose quotient is being analysed and the target of the membership statements.
- The torsion-free criterion identifying modules that have no I-torsion. invoked at proof lines 24063-24065. Turns the two-sided torsion statement into the vanishing of M/M[I^infinity].

Source reference. Original lemma 24024: sources/stacks-more-algebra/pinned-original-source.tex, source lines 24024–24024. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *Let R be a ring. Let I be a finitely generated ideal of R.*

(1) *For any R-module M we have $(M/M[I^\infty])[I] = 0$.*
(2) *An extension of I-power torsion modules is I-power torsion.*

Proof. Let $m \in M$. If m maps to an element of $(M/M[I^\infty])[I]$ then $Im \subset M[I^\infty]$. Write $I = (f_1, \ldots, f_t)$. Then we see that $f_i m \in M[I^\infty]$, i.e., $I^{n_i} f_i m = 0$ for some $n_i > 0$. Thus we see that $I^N m = 0$ with $N = \sum n_i + 2$. Hence m maps to zero in $(M/M[I^\infty])$ which proves the first statement of the lemma.

For the second, suppose that $0 \to M' \to M \to M'' \to 0$ is a short exact sequence of modules with M' and M'' both I-power torsion modules. Then $M[I^\infty] \supset M'$ and hence $M/M[I^\infty]$ is a quotient of M'' and therefore I-power torsion. Combined with the first statement and Lemma 24024 this implies that it is zero. □

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